



Figure 3: Incoming call at user-Y (709903)

recorded voice message, “Please press any key to accept a call for extension” but hangs up on the call, so a notification goes back to the Asterisk server.

6. Asterisk routes the call to the support manager and finally the client’s call is established.
7. If the support manager denies the call, it is redirected to the primary on-call person’s voice mail.

The *FollowMe* feature redirects the calls one after the other until either someone answers the call or till the end of the dial plan.

Smartphone support for Android, iPhone and BlackBerry

Asterisk supports softphones that can be installed on smartphones running on Android, as well as the iPhone and BlackBerry. It is a key feature for smartphone users with 3G, making communication easier, simpler and cheaper. It also increases the resource reachability—if the resource is not at the desk to pick the phone, the call can be diverted to his smartphone or Gtalk.

Text-to-speech with Festival

Festival is another open source speech synthesiser. Asterisk can be integrated with *Festival* to process text-to-speech conversion in auto attendants or interactive response systems. *Festival* is available for many major languages like English, Spanish and even some Indian regional languages like Hindi, Telugu and it enables the automation of tasks.

Voicemail

Unlike Cisco, Asterisk does not use Exchange Server to manage voice mails. Cisco manages voice mails as separate entities with Microsoft Exchange Server, which increases the cost. Asterisk uses its own PBX engine and voice mails can be delivered at the email address of the user, which can be saved and retrieved any time.

Database connections

Asterisk can be integrated with databases to fetch and process the data required in Interactive Voice Response Systems (IVRS). IVRS can be implemented using tools like *Festival* which can read the data from the database. For example, if a bank has IVRS in place, a customer can call the bank and get the details of his fund balance by IVRS. Asterisk can implement a low cost IVRS system.

Hobby telephone systems

In this section, let’s look at how to build a hobby telephone system with a real-time approach by converting an ordinary system to a PBX communications server. This approach can be used on a bigger scale for real-world purposes.

Installation

Asterisk can turn your computer into a communications server in a few minutes. It can be easily installed and configured to run VoIP telephony from scratch, without any special hardware requirements.

The minimum hardware requirements for a small home PBX network are mentioned below.

Processor: PII – 400MHz/ AMD 2400+/ AMD Sempron 3200+
 RAM: 512 MB +
 Hard disk: 4 GB
 PCI card: FXO/FXS card (optional - only for connecting analogue phones)

Asterisk is available as a software download and also as a Live CD called ‘Asterisk Now’. Choose either, based on your choice and comfort.

Asterisk (software download)

This can be installed on any machine that has a working Linux operating system. It is supported by all major distributions of Linux like Debian, Ubuntu, Centos and Red Hat. It can be